

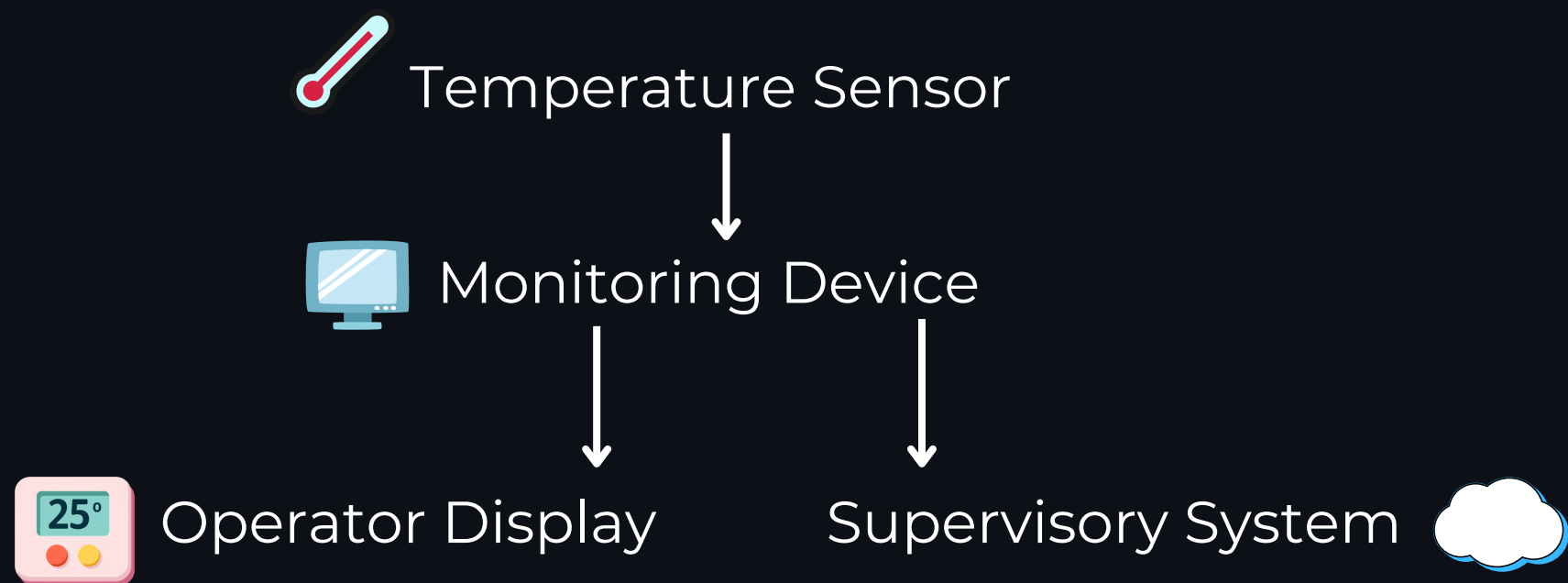
Reliability Technique #2 - Retry

One Sensor Read Failed. Should the System Fail Too?

Using a Temperature Monitoring Device as an example



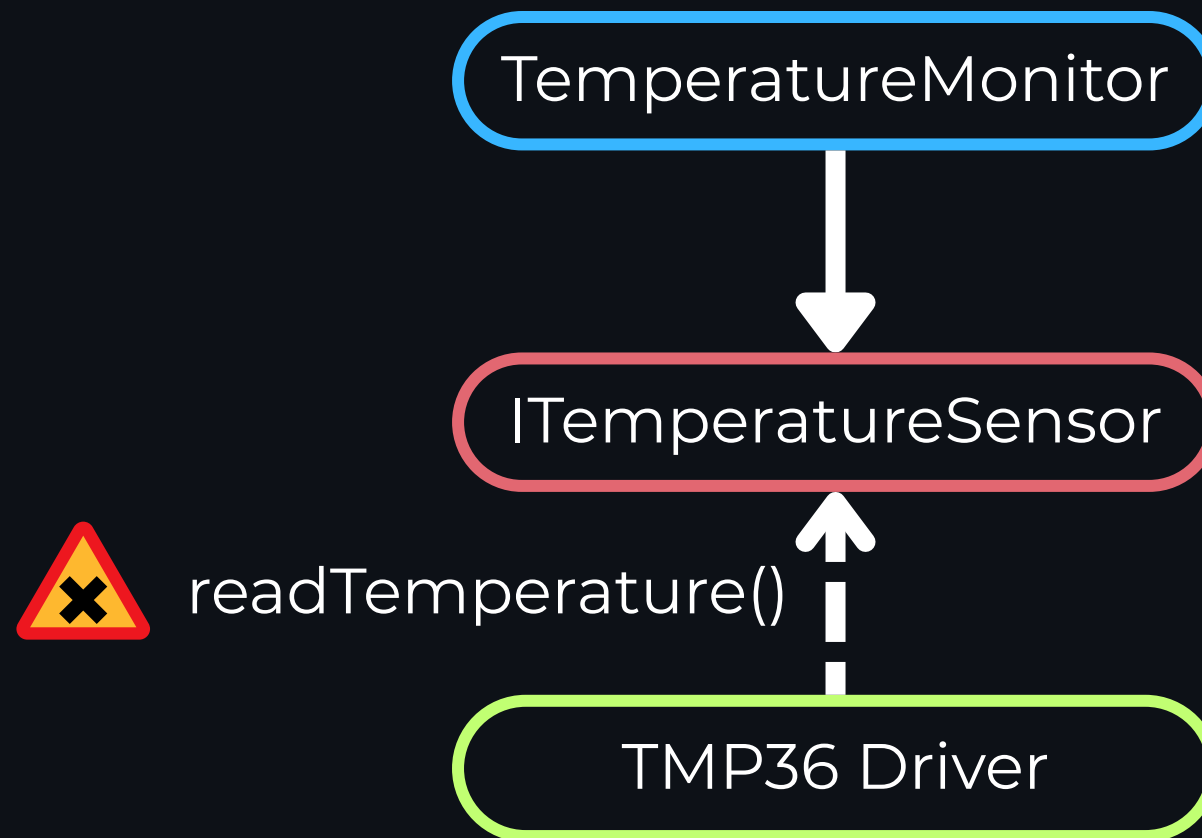
The System



Responsibilities:

- ▶ Read sensor
- ▶ Display temperature
- ▶ Report measurement

Our HAL Solved One Problem



- ▶ Application is isolated from the specific sensor

But abstraction doesn't make hardware failures disappear

What happens when `readTemperature()` temporarily fails?

Not Every Failure Is Permanent

A sensor read can fail for a transient reason.

Examples:

- ▶ Temporary communication error/timeout
- ▶ Bus contention, electrical noise, sensor busy



Should one transient failure stop monitoring?

Two Extremes — Both Can Be Wrong

A failed operation needs a deliberate recovery strategy

Naive reactions:

- ▶ Read fails → System Error
- ▶ Retry forever → System may block indefinitely

Neither distinguishes transient from persistent faults

Recovery needs a policy.

Introduce a Bounded Retry Policy

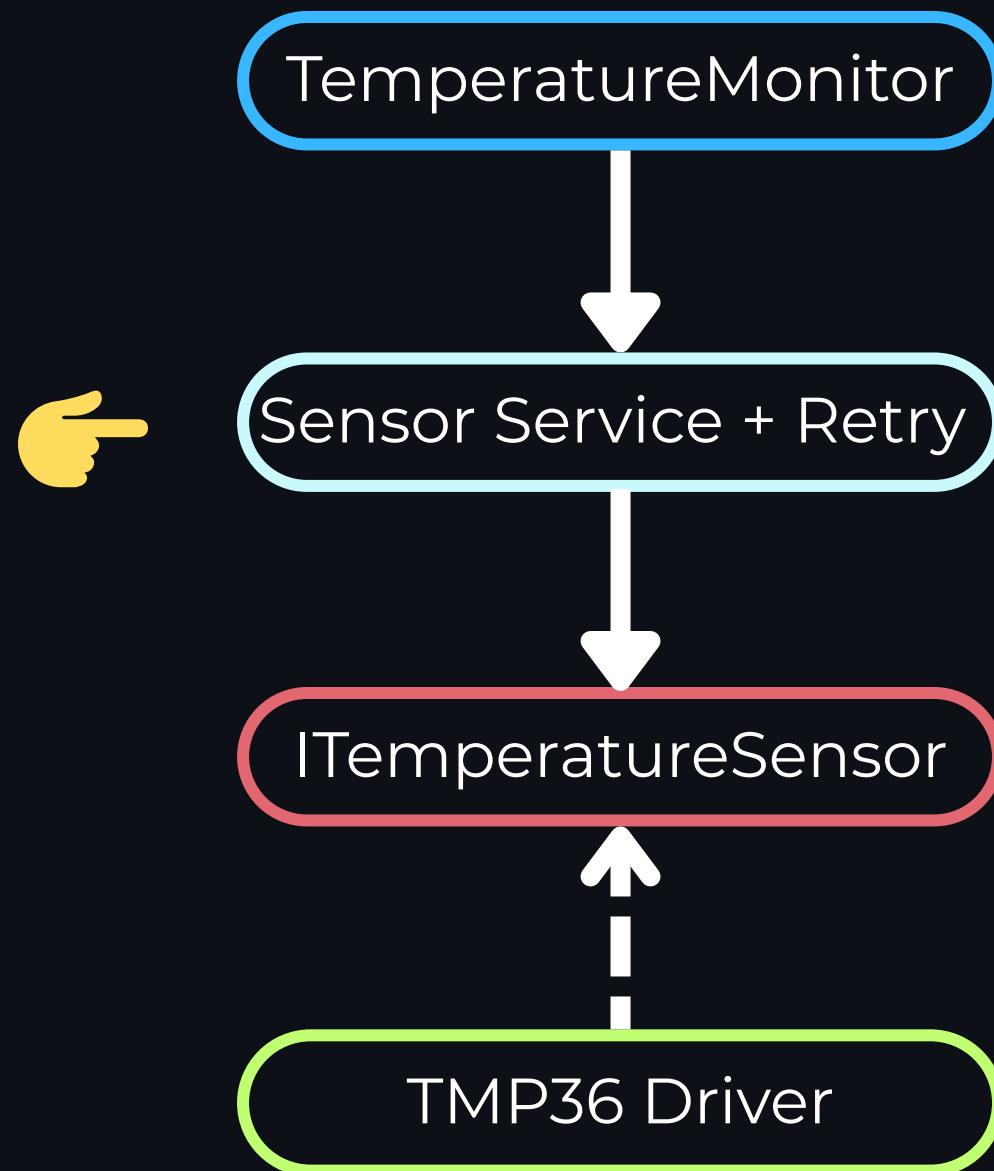
Retry only when the failure may be temporary

Architecture questions:

- ▶ How many retries? How long between attempts?
- ▶ Which failures are actually retryable?
- ▶ What happens when retries are exhausted?

Read → Fail → Wait → Retry → Escalate if exhausted

Where Should Retry Live?



Business logic “TemperatureMonitor” is separated from recovery policy.

Retry Improves Resilience — But Isn't Free

Retry only when the failure may be temporary

Benefits:

- ✓ Recovers from transient faults
- ✓ Prevents unnecessary system failures

Tradeoffs:

- ⚠ Increases response latency
- ⚠ Can hide persistent faults if badly designed

Retry is a policy — not simply a loop.

Takeaway:

Retry only when the failure may be temporary

Don't Ask:

Should we use Retry?

Ask:

Which failures are transient? How much recovery is safe?

Good Retry design: Selective → Bounded → Observable → Escalated

Want the Full Design?

GitHub contains:

- ✓ UML diagram
- ✓ C++ source code
- ✓ Design trade-offs

Repository link in comments

Coming Next — Watchdog....

What if the problem isn't a failed sensor read.....

But software that stops responding completely?